SIVA KUMAR ANNAM | DevOps & Cloud Engineer

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PROFESSIONAL SUMMARY

DevOps and Cloud-oriented Engineer with 1.5+ years of experience in automation, infrastructure, and service assurance platforms. Hands-on with Python/Bash automation, AWS services, Docker, Kubernetes, Terraform, and CI/CD pipelines. Built monitoring solutions, reduced manual operational effort by ~30%, and delivered GitOps-style deployment workflows. Additional hands-on experience building a RAG chatbot with LangChain and Ollama. Strong foundation in Linux, networking, observability (Prometheus/Grafana), and continuous delivery practices.

WORK EXPERIENCE

Associate Software Engineer | Automation & Platform Engineer

MasterCom Technology Services India Pvt Ltd, Bangalore | Nov 2024 – Present (~1.5 years)

Project: SevOne NMS & Service Assurance Automation | Client: Tata Communications Ltd (Pune)

- Automated network management and operational tasks using Python, Perl and Shell scripts (device discovery, backups, log cleanup, performance checks) — reduced manual effort by ~30%.
- Worked on SevOne NMS, PMS and DISHA (Assure1) for proactive/reactive monitoring; created dashboards, alerts, thresholds and performance reports.
- Implemented monitoring and observability practices; analyzed metrics via SNMP, NetFlow and logs; supported RCA and high-availability service assurance.
- Performed MySQL → PostgreSQL database migrations using Python/Perl (table mapping, join rewrites, testing of jobs/services); used DBeaver for administration.
- Developed Flask-based RESTful APIs with caching and performance optimization (reduced response times and database queries).
- Provided technical support for Assure1 & Spotlight: configuration, real-time monitoring, troubleshooting, and RCA for network/service incidents.
- Collaborated on service assurance operations, ticket/incident management via ServiceNow, and SLA compliance.
- Actively applied DevOps practices: Infrastructure as Code concepts, containerization, CI/CD and GitOps-style workflows; version-controlled scripts in Git.

Cloud, Containers & CI/CD (Hands-on / Project Experience)

- Worked with AWS services: EC2, S3, RDS, VPC, IAM, EBS, ELB, Route 53, Auto Scaling, CloudWatch and Lambda.
- Implemented Infrastructure as Code (IaC) patterns with Terraform (modules, backends) and AWS CloudFormation concepts.
- Worked with Docker for containerized applications and Kubernetes for orchestration (resource limits, ConfigMaps, Secrets).
- Built CI/CD pipelines using GitHub Actions, Jenkins, GitLab CI/CD and AWS CodePipeline/CodeBuild/CodeDeploy concepts; integrated Nexus, SonarQube and Slack.
- Set up monitoring stacks with Prometheus, Grafana, Loki and CloudWatch for observability.

PROJECTS

1. SevOne NMS & Service Assurance Automation (Client: Tata Communications Ltd)

Technologies: Python, Perl, Bash, Flask, MySQL, PostgreSQL, SevOne NMS, DISHA (Assure1), Linux, ServiceNow

- End-to-end automation of monitoring, reporting and operational tasks on the SevOne platform for a large telecom client.
- Designed modular Python/Perl scripts that collect metrics, generate reports and trigger alerts, significantly cutting manual effort.
- Migrated critical database workloads from MySQL to PostgreSQL with careful join rewrites and validation.
- Built Flask APIs to expose processed data and support internal tooling needs; supported Assure1/Spotlight for configuration, monitoring and RCA.

2. RAG Chatbot using LangChain + Ollama (Hands-on Generative AI Project)

Technologies: Python, LangChain, Ollama, LLMs, Embeddings, RAG, Prompt Engineering | GitHub: github.com/Sivakumarannam/rag-chatbot-langchain-ollama

- Built an end-to-end Retrieval-Augmented Generation (RAG) chatbot using LangChain and Ollama (local LLMs).
- Implemented document loading, text splitting, embedding generation, and vector store for context-aware answers.
- Designed and tested prompt engineering strategies to improve response quality and reduce hallucinations.
- Created a practical chatbot that answers questions based on custom knowledge base (PDF/text documents).
- Explored key Generative AI concepts: LLMs, Embeddings, Fine-tuning, RAG, Prompt Engineering and Agentic AI patterns.
- Set up local development environment, integrated ChatGPT API for comparison, and documented the full workflow.
- Published the complete project on GitHub: https://github.com/Sivakumarannam/rag-chatbot-langchain-ollama.git

TECHNICAL SKILLS

Programming & Scripting: Python (strong), Perl, Bash, Shell Scripting, JavaScript, PL/SQL
Web & Frameworks: Flask, Django (basic knowledge)
Generative AI / LLM: LangChain, Ollama, RAG, Prompt Engineering, Embeddings, ChatGPT API, LLMs
Cloud & DevOps: AWS (EC2, S3, RDS, VPC, IAM, Lambda, CloudWatch, CodePipeline, EKS, ECS), Terraform, Ansible, Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, Nexus, SonarQube
Monitoring & Observability: SevOne NMS, PMS, DISHA (Assure1), Prometheus, Grafana, CloudWatch
Databases: MySQL, PostgreSQL, DBeaver — CRUD operations, migrations, query optimization
Networking & Protocols: SNMP, NetFlow, performance and alert log analysis, Linux/Unix administration
Version Control & ITSM: Git, ServiceNow, GitOps practices
Other: Pytest, VS Code, PuTTY, WinSCP, writing clean and maintainable code

KEY LEARNING & SELF-PROJECTS

GitOps Deployment Platform (Capstone) — Built an end-to-end GitOps-style platform using GitHub Actions and Kubernetes for declarative continuous delivery. Focused on pipeline design, environment promotion and automated deployments.

DevOps & Cloud Mastery (Self-Paced) — Linux & server management, Networking fundamentals, YAML/JSON/Bash, AWS core services (IAM, EC2, S3, RDS, VPC, Lambda, Auto Scaling, Route 53, Systems Manager), Terraform modules & backends, Ansible, Prometheus/Grafana/Loki, Docker & production-grade Kubernetes, full CI/CD with Jenkins/GitHub Actions/GitLab, Nexus, SonarQube, Slack integration. Python scripting for automation and AWS tasks.

Generative AI Learning Path — Structured modules on Generative AI fundamentals, LLMs, Prompt Engineering, Embeddings, Fine-tuning, RAG, Agentic AI, industry use-cases, ethical considerations, and practical chatbot development with ChatGPT API and local LLMs (Ollama).

CERTIFICATIONS

- Python Full Stack Development — 2023
- Full Stack Development Internship — Innomatics Research Labs, 2023

EDUCATION

Bachelor of Technology in Computer Science and Engineering
St. Mary's Group of Institutions, Guntur, Andhra Pradesh | 2019 – 2023 | CGPA – 6.8

Intermediate (Class 12 / +2)
NRI Junior College, Tenali, Andhra Pradesh| 2017 – 2019 | CGPA: 8.7

Secondary School Certificate (Class 10)
Nehru Niketan English Medium High School |Tenali, Andhra Pradesh| CGPA: 8.7

LANGUAGES

English, Telugu (Mother Tongue), Hindi

STRENGTHS

- Strong analytical and problem-solving skills in network and service issues
- Quick learner with keen interest in Generative AI, RPA and scalable application development
- Good collaboration with cross-functional teams and mentoring of juniors
- Focus on SLA compliance, documentation and knowledge sharing